

Course 4131

4 Credits: 4

Program Core

Source-grounded

Object Oriented Programming

- Understand object-oriented Principles.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 4131 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4131.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering / Computer Hardware Engineering
Course code	4131
Course title	Object Oriented Programming
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

31 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Understand object-oriented Principles.
- Write, Compile and Execute Java Programs.
- Familiarize GUI programming using Swing.
- Introduce database connectivity for developing applications.

Course outcomes

CO	Official outcome	Study level
CO1	16 Applying develop simple JAVA applications. Build classes using inheritance, interfaces and	See official syllabus
CO2	16 Applying packages	See official syllabus
CO3	Build GUI applications using swing. 14 Applying	See official syllabus
CO4	Develop applications using Database. 12 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	applications	12	As specified
Module 2	Build classes using inheritance, interfaces and packages.	8	As specified
Module 3	Build GUI applications using swing.	6	As specified
Module 4	Develop applications using database	5	As specified

MODULE 1

applications

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: applications
- Official duration: As specified
- Official topic entries captured: 12

- The full source excerpt is preserved below for coverage checking.

1 Understanding. Programming. Explain the JDK & IDE's for developing java

1 Understanding. Programming. Explain the JDK & IDE's for developing java is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. Programming. Explain the JDK & IDE's for developing java using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. programming. explain the jdk & ide's for developing java should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. Programming. Explain the JDK & IDE's for developing java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- 1 Understanding. Programming. Explain the JDK & IDE's for developing java
 definition/meaning
 .
 , steps, application
 . Technical terms English-
 .

1 Understanding. applications Outline the general structure of a Java Understanding.

1 Understanding. applications Outline the general structure of a Java Understanding. is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. applications Outline the general structure of a Java Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. applications outline the general structure of a java understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. applications Outline the general structure of a Java Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. applications Outline the general structure of a Java Understanding. ഓരോന്നിന്റെയും definition/meaning ഉൾപ്പെടുത്തുക. ഓരോന്നും ഏതു രീതിയിൽ, steps, application എന്നിവ ഉൾപ്പെടുത്തുക. Technical terms English-ൽ ഉൾപ്പെടുത്തുക.

program and explain steps to create, compile 1 and execute a program

Describe objects and classes and use classes

program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, program and explain steps to create, compile 1 and execute a program describe objects and classes and use classes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രോഗ്രാം എഴുതുകയും പ്രവർത്തിപ്പിക്കുകയും ചെയ്യുക. വസ്തുക്കളെക്കുറിച്ച് വിവരിക്കുകയും ക്ലാസ്സുകൾ ഉപയോഗിക്കുകയും ചെയ്യുക. വസ്തുക്കളുടെ നിർവ്വചനം/അർത്ഥം, പ്രവർത്തിപ്പിക്കൽ, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ച് വിവരിക്കുക. Technical terms English-ൽ എഴുതുക.

2 Understanding. to model objects. Demonstrate how to define classes and create

2 Understanding. to model objects. Demonstrate how to define classes and create is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. to model objects. Demonstrate how to define classes and create using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. to model objects. demonstrate how to define classes and create should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. to model objects. Demonstrate how to define classes and create means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding. to model objects.

Demonstrate how to define classes and create **class** definition/meaning **class**. **class** **class** **class**, steps, application **class** **class** **class**. **Technical terms** English-**class** **class** **class**.

1 Applying. objects

1 Applying. objects is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying. objects using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying. objects should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying. objects means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-ഓ 1 Applying. objects ഓരോന്നിനും ഓരോന്നിനും definition/meaning ഓരോന്നിനും. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നിനും ഓരോന്നും. Technical terms English-ഓ ഓരോന്നും ഓരോന്നിനും.

Create Objects using constructors. 1 Applying Interpret accessing objects via object

Create Objects using constructors. 1 Applying Interpret accessing objects via object is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Create Objects using constructors. 1 Applying Interpret accessing objects via object using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, create objects using constructors. 1 applying interpret accessing objects via object should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Create Objects using constructors. 1 Applying Interpret accessing objects via object means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-ഓജക്ട് Create Objects using constructors. 1 Applying Interpret accessing objects via object ഓജക്ട് definition/meaning ഓജക്ട്. ഓജക്ട് ഓജക്ട് ഓജക്ട്, steps, application ഓജക്ട് ഓജക്ട്. Technical terms English-ഓജക്ട് ഓജക്ട്.

1 Understanding. reference variables. Distinguish between instance and static

1 Understanding. reference variables. Distinguish between instance and static is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. reference variables. Distinguish between instance and static using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. reference variables. distinguish between instance and static should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. reference variables. Distinguish between instance and static means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാഠ്യ 1 Understanding. reference variables.

Distinguish between instance and static പദപ്രയോഗം പദപ്രയോഗം definition/meaning പദപ്രയോഗം. പദപ്രയോഗം പദപ്രയോഗം, steps, application പദപ്രയോഗം പദപ്രയോഗം പദപ്രയോഗം. Technical terms English-പദപ്രയോഗം പദപ്രയോഗം.

1 Understanding. variables and methods. Develop Java Programs to implement classes

1 Understanding. variables and methods. Develop Java Programs to implement classes is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. variables and methods. Develop Java Programs to implement classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. variables and methods. develop java programs to implement classes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. variables and methods. Develop Java Programs to implement classes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാഠ്യ 1 Understanding. variables and methods. Develop Java Programs to implement classes പാഠ്യപരിചയം പാഠ്യപരിചയം definition/meaning പാഠ്യപരിചയം. പാഠ്യപരിചയം പാഠ്യപരിചയം, steps, application പാഠ്യപരിചയം പാഠ്യപരിചയം പാഠ്യപരിചയം. Technical terms English-പാഠ്യ പാഠ്യ പാഠ്യപരിചയം.

2 Applying. and objects. Explain Visibility Modifiers and Data

2 Applying. and objects. Explain Visibility Modifiers and Data is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying. and objects. Explain Visibility Modifiers and Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying. and objects. explain visibility modifiers and data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying. and objects. Explain Visibility Modifiers and Data means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-2 Applying. and objects. Explain Visibility Modifiers and Data definition/meaning. steps, application Technical terms English- .

1 Understanding. Encapsulation Develop methods with object arguments an

1 Understanding. Encapsulation Develop methods with object arguments an is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. Encapsulation Develop methods with object arguments an using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. encapsulation develop methods with object arguments an should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. Encapsulation Develop methods with object arguments an means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-1 Understanding. Encapsulation Develop methods with object arguments an ഏകീകൃതതയ്ക്ക് നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-1 നിർവ്വചനം നിർവ്വചിക്കുക.

differentiate between Primitive arguments 2 Applying and object type arguments

differentiate between Primitive arguments 2 Applying and object type arguments is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify differentiate between Primitive arguments 2 Applying and object type arguments using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, differentiate between primitive arguments 2 applying and object type arguments should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What differentiate between Primitive arguments 2 Applying and object type arguments means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\frac{1}{2}$ differentiate between Primitive arguments 2 Applying and object type arguments $\frac{1}{2}$ definition/meaning $\frac{1}{2}$. $\frac{1}{2}$ steps, application $\frac{1}{2}$. Technical terms English- $\frac{1}{2}$.

Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) - Characteristics of OOP - Features of JAVA - Advantages of JAVA - Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) - Building Java applications. Objects and Classes - Defining a Class - Declaring attributes, Declaring and Defining methods, Creating Object - Accessing Objects - Constructors - Constructor overloading - Static variables, constants and methods - method overloading - Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement.

Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) - Characteristics of OOP - Features of JAVA - Advantages of JAVA - Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) - Building Java applications. Objects and Classes - Defining a Class - Declaring attributes, Declaring and Defining methods, Creating Object - Accessing Objects - Constructors - Constructor overloading - Static variables, constants and methods - method overloading - Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) - Characteristics of OOP - Features of JAVA - Advantages of JAVA - Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) - Building Java applications. Objects and Classes - Defining a Class - Declaring attributes, Declaring and Defining methods, Creating Object - Accessing Objects - Constructors - Constructor overloading - Static variables, constants and methods - method overloading - Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain exception handling. 2 understanding contnets: object oriented programming (oop) – characteristics of oop – features of java – advantages of java – tools available of java programming (jdk, java packages, various ides like netbeans, eclipse) – building java applications. objects and classes – defining a class – declaring attributes, declaring and defining methods, creating object – accessing objects - constructors – constructor overloading – static variables, constants and methods – method overloading – visibility modifiers, data field encapsulation, passing and returning objects as arguments, array of objects-exception handling, try,catch,- multiple catch & finally statement. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects

and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. **പ്രധാനപ്പെട്ട പദങ്ങളുടെ definition/meaning നൽകുക.** **പ്രധാനപ്പെട്ട പദങ്ങളുടെ ഉപയോഗം, steps, application നൽകുക.** **Technical terms English-ൽ നൽകുക.**

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

applications			
	Explain the features of Object-Oriented		
M1.01		1	
Understanding.			
	Programming.		
	Explain the JDK & IDE’s for developing java		
M1.02		1	
Understanding.			
	applications		
	Outline the general structure of a Java		
Understanding.			
M1.03	program and explain steps to create, compile and execute a program	1	
	Describe objects and classes and use classes		
M1.04		2	
Understanding.			
	to model objects.		
	Demonstrate how to define classes and create		
M1.05		1	Applying.
	objects		
M1.06	Create Objects using constructors.	1	Applying
	Interpret accessing objects via object		
M1.07		1	
Understanding.			
	reference variables.		
	Distinguish between instance and static		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 2

Build classes using inheritance, interfaces and packages.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build classes using inheritance, interfaces and packages.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

3 Understanding inheritance with examples. Illustrate invoking the superclass constructor

3 Understanding inheritance with examples. Illustrate invoking the superclass constructor is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding inheritance with examples. illustrate invoking the superclass constructor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding inheritance with examples. Illustrate invoking the superclass constructor $\overline{\text{പ്രൊട്ടോടൈപ്പ്}} \overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ definition/meaning $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$. $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$, steps, application $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$. Technical terms English- $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$ $\overline{\text{പ്രൊട്ടോടൈപ്പ്}}$.

2 Applying and methods using super keywords. Illustrate overriding instance methods in the

2 Applying and methods using super keywords. Illustrate overriding instance methods in the is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying and methods using super keywords. Illustrate overriding instance methods in the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying and methods using super keywords. illustrate overriding instance methods in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying and methods using super keywords. Illustrate overriding instance methods in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying and methods using super keywords.
Illustrate overriding instance methods in the Python class definition/meaning. Explain the steps, application and technical terms English-Malayalam.

2 Applying subclass.

2 Applying subclass. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying subclass. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying subclass. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying subclass. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying subclass. definition/meaning . steps, application . Technical terms English-

Illustrate polymorphism and dynamic binding

Illustrate polymorphism and dynamic binding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate polymorphism and dynamic binding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate polymorphism and dynamic binding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate polymorphism and dynamic binding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓരോ ഉദാഹരണം പോലും define/meaning ഉൾപ്പെടുത്തണം. ഓരോ ഉദാഹരണം പോലും, steps, application ഉൾപ്പെടുത്തണം ഉൾപ്പെടണം. Technical terms English-ൽ ഉൾപ്പെടുത്തണം.

Explain visibility controls 1 Understanding Build abstract class, final class and method

Explain visibility controls 1 Understanding Build abstract class, final class and method is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain visibility controls 1 Understanding Build abstract class, final class and method using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain visibility controls 1 understanding build abstract class, final class and method should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain visibility controls 1 Understanding Build abstract class, final class and method means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നിരിക്കട്ടെ Explain visibility controls 1 Understanding Build abstract class, final class and method എന്നിവയുടെ definition/meaning എന്നിവയെക്കുറിച്ചുള്ള. എന്നിവയെക്കുറിച്ചുള്ള definition, steps, application എന്നിവയെക്കുറിച്ചുള്ള. Technical terms English-എന്നിരിക്കട്ടെ എന്നിവയെക്കുറിച്ചുള്ള.

2 Applying overriding

2 Applying overriding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying overriding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying overriding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying overriding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying overriding തർക്കപ്പെടുന്ന നിർവ്വചനം/അർത്ഥം തർക്കപ്പെടുന്ന. തർക്കപ്പെടുന്ന തർക്കപ്പെടുന്ന, steps, application തർക്കപ്പെടുന്ന തർക്കപ്പെടുന്ന. Technical terms English- തർക്കപ്പെടുന്ന തർക്കപ്പെടുന്ന.

Develop interfaces

Develop interfaces is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop interfaces using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop interfaces should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop interfaces means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നു വികസിപ്പിക്കുക ഇറക്കുമതി ചെയ്യുക
definition/meaning നിർവ്വചനം/അർത്ഥം. നിർവ്വചനം നിർവ്വചനം, steps, application
നിയമങ്ങൾ നിർവ്വചനങ്ങൾ നിർവ്വചനം. Technical terms English-എന്നു വികസിപ്പിക്കുക.

Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages.

Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop packages 2 applying inheritance - types of inheritance - superclass and subclass - calling super class constructor and methods - overloading vs overriding - final variables, final methods and final classes - abstract methods and class - polymorphism - dynamic binding - visibility controls. interfaces - definition -

extending, implementing and accessing interfaces- packages – creating and accessing of packages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding – Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding – Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Build classes using inheritance, interfaces and packages.

Describe inheritance and different types of

M2.01		3	
Understanding			
	inheritance with examples.		
	Illustrate invoking the superclass constructor		
M2.02		2	Applying
	and methods using super keywords.		
	Illustrate overriding instance methods in the		
M2.03		2	Applying
	subclass.		
M2.04	Illustrate polymorphism and dynamic binding	2	Applying
M2.05	Explain visibility controls	1	
Understanding			
	Build abstract class, final class and method		
M2.06		2	Applying
	overriding		
M2.07	Develop interfaces	2	Applying
M2.08	Develop packages	2	Applying
	Series Test I	1	

Contents:

Inheritance – Types of Inheritance – Superclass and subclass – calling super class

constructor and methods – overloading vs overriding - Final variables, final methods and

final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Build GUI applications using swing.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build GUI applications using swing.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI

Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline gui programming in java using swing. 2 understanding. make use of packages containing gui should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓൗട്ട്ലൈൻ GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI ഓൗട്ട്ലൈൻ GUI definition/meaning ഓൗട്ട്ലൈൻ GUI. ഓൗട്ട്ലൈൻ GUI, steps, application ഓൗട്ട്ലൈൻ GUI. Technical terms English-ഓൗട്ട്ലൈൻ GUI.

containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI

containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, containers, components, event-handling classes 2 applying. and interfaces to develop simple applications. describe steps to create and manipulate gui should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI
 definition/meaning
 .
 , steps, application
 . Technical terms English-
 .

3 Understanding. controls. Explain handling of mouse events and

3 Understanding. controls. Explain handling of mouse events and is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. controls. Explain handling of mouse events and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. controls. explain handling of mouse events and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. controls. Explain handling of mouse events and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 3 Understanding. controls. Explain handling of mouse events and definition/meaning .
 , steps, application . Technical terms English- .

2 Understanding. keyboard events. Explain Common GUI event type and Listener

2 Understanding. keyboard events. Explain Common GUI event type and Listener is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. keyboard events. Explain Common GUI event type and Listener using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. keyboard events. explain common gui event type and listener should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. keyboard events. Explain Common GUI event type and Listener means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-2 Understanding. keyboard events. Explain Common GUI event type and Listener definition/meaning. steps, application. Technical terms English-

2 Understanding. types.

2 Understanding. types. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. types. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-2 Understanding. types. definition/meaning . steps, application . Technical terms English-

Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers.

Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using swing components 3 applying. outline gui programming - swing - swing components - simple gui based input/output programs - gui application development using swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - event handling - common gui event types and containers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextarea – Event Handling – Common GUI event Types and Containers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextarea – Event Handling – Common GUI event Types and Containers.

മലയാളത്തിൽ നിർവ്വചന/അർത്ഥം നൽകുക. ഘട്ടങ്ങൾ, ഘട്ടങ്ങൾ, ഘട്ടങ്ങൾ, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Build GUI applications using swing.

M3.01	Outline GUI Programming in Java using swing.	2
Understanding.		
	Make use of packages containing GUI	
M3.02	containers, components, event-handling classes	2
Applying.		
	and interfaces to develop simple applications.	
	Describe steps to create and manipulate GUI	
M3.03		3
Understanding.		
	controls.	
	Explain handling of mouse events and	
M3.04		2
Understanding.		
	keyboard events.	
	Explain Common GUI event type and Listener	
M3.05		2
Understanding.		
	types.	
M3.06	Develop programs using swing Components	3
Applying.		

Contents:
Outline GUI programming – Swing – Swing Components – Simple GUI based
input/output programs - GUI application development using Swing components such
as
Jlabel, JTextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea –

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Develop applications using database

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop applications using database
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Explain relational database & SQL

Explain relational database & SQL is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain relational database & SQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain relational database & sql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain relational database & SQL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain relational database & SQL

relational database definition/meaning relational database. relational database steps, application relational database steps. Technical terms English- relational database.

Explain SQL commands.

Explain SQL commands. is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain SQL commands. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sql commands. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain SQL commands. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain SQL commands. SQL commands definition/meaning. SQL commands steps, application. Technical terms English- Explain SQL commands.

Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the

Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the stages in jdbc 2 understanding illustrate the interfaces in jdbc -register the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഈ Illustrate the stages in JDBC 2 Understanding
Illustrate the interfaces in JDBC -register the ഏകദേശം ഏകദേശം
definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application
ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഈ ഏകദേശം ഏകദേശം.

driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset

driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, driver- connection- preparedstatement- 2 applying executing statements- resultset should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഡ്രൈവർ- കണക്കിനെ- Preparedstatement- 2
Applying Executing statements- Resultset ഫലസെറ്റ് definition/meaning
ഫലസെറ്റ് ഫലസെറ്റ്, steps, application ഫലസെറ്റ് ഫലസെറ്റ്
ഫലസെറ്റ്. Technical terms English-ഡ്രൈവർ ഫലസെറ്റ്.

Develop simple application with database 4 Applying Database - Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity - JDBC - RowSet interface - PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book

Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A

Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java,

6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson,

9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3

Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java

How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources

Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2

<https://www.tutorialspoint.com/java>

Develop simple application with database 4 Applying Database - Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity - JDBC - RowSet interface - PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java> is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop simple application with database 4 Applying Database - Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity - JDBC - RowSet interface - PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th

edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1

https://onlinecourses.nptel.ac.in/noc20_cs08/course 2

<https://www.tutorialspoint.com/java> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop simple application with database 4 applying database – creation of databases-tables -sql commands like -create, alter, drop, select, insert, delete database connectivity – jdbc – rowset interface – preparedstatement - retrieving results - closing connection. text / reference t/r book title/author herbert schildt, dale skrien: java fundamentals a comprehensive t1 introduction t2 balaguruswamy e: programming with java, 6th edition. r1 liang, y daniel: introduction to java programming, pearson, 9th ed. r2 herbert schildt: java the complete reference, seventh edition, r3 herbert schildt: swing a beginner's guide r4 paul deital, harvey deital: java how to program, r5 yang hu: easy learning jdbc+mysql online resources sl.no website link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Develop simple application with database 4
Applying Database – Creation of databases-Tables -SQL commands like -create,
alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface –
PreparedStatement - retrieving results - closing connection. Text / Reference T/R
Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A
Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th
edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2
Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt:
Swing A Beginner’s guide R4 Paul Deital, Harvey Deital: Java How to Program, R5
Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1
https://onlinecourses.nptel.ac.in/noc20_cs08/course 2
<https://www.tutorialspoint.com/java> □□□□□□□□□□ □□□□□ definition/meaning
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□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Develop applications using database			
M4.01	Explain relational database & SQL	2	
Understanding			
M4.02	Explain SQL commands.	2	
Understanding			
M4.03	Illustrate the stages in JDBC	2	
Understanding			
M4.04	Illustrate the interfaces in JDBC –register the driver- connection- PreparedStatement-	2	
Applying			
	Executing statements- Resultset		
M4.05	Develop simple application with database	4	Applying
	Series Test – II	1	

Contents:

Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection.

Text / Reference

T/R	Book Title/Author
T1	Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive Introduction

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding. Programming. Explain the JDK & IDE's for developing java. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. Programming. Explain the JDK & IDE's for developing java*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding. applications Outline the general structure of a Java Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. applications Outline the general structure of a Java Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes. (3 marks)

Answer: Begin with the standard definition or identity of *program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding. to model objects. Demonstrate how to define classes and create. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. to model objects. Demonstrate how to define classes and create.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding inheritance with examples. Illustrate invoking the superclass constructor*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Applying and methods using super keywords. Illustrate overriding instance methods in the. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying and methods using super keywords. Illustrate overriding instance methods in the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Applying subclass.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying subclass..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate polymorphism and dynamic binding. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate polymorphism and dynamic binding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 3

Define or explain Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI. (3 marks)

Answer: Begin with the standard definition or identity of *Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI. (3 marks)

Answer: Begin with the standard definition or identity of *containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. Describe steps to create and manipulate GUI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding. controls. Explain handling of mouse events and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. controls. Explain handling of mouse events and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding. keyboard events. Explain Common GUI event type and Listener. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. keyboard events. Explain Common GUI event type and Listener.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Explain relational database & SQL. (3 marks)

Answer: Begin with the standard definition or identity of *Explain relational database & SQL.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain SQL commands.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain SQL commands..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset. (3 marks)

Answer: Begin with the standard definition or identity of *driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding. Programming. Explain the JDK & IDE's for developing java.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding inheritance with examples. Illustrate invoking the superclass constructor.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain relational database & SQL**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.

Term	Meaning in this lesson
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://onlinecourses.nptel.ac.in/noc20_cs08/course <https://www.tutorialspoint.com/java>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 4131. Verify institutional updates against the current official curriculum.